

- 3 (a) Oscillatory motion where the acceleration is proportional to the displacement from the equilibrium position, and directed towards the equilibrium position. [1]
- (b) (i) $mg = kx \Rightarrow k = \frac{mg}{x} = \frac{0.25 \times 9.81}{0.040} = 61.3 \text{ N m}^{-1}$ [1]
- (ii) 1. $a = -\frac{kx}{m}$
 $\omega^2 = \frac{k}{m}$ [1]
 $2\pi f = \sqrt{\frac{k}{m}}$
 $f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$
 $= \frac{1}{2\pi} \sqrt{\frac{61.3}{0.69}}$ [1]
 $= 1.5 \text{ Hz}$ [0]
2. PE = KE
 $\frac{1}{2}m\omega^2x^2 = \frac{1}{2}m\omega^2(x_0^2 - x^2)$ [1]
 $x^2 = x_0^2 - x^2$
 $2x^2 = 0.020^2$
 $x = 0.0141 \text{ m}$
 $= 14.1 \text{ mm}$ [1]
3. Increasing amplitude from 0.2 Hz to 1.5 Hz [1]
Decreasing amplitude from 1.5 Hz to 3.0 Hz [1]
Maximum energy transfer when driver frequency is equal to spring-mass's natural frequency [1]
Hence, maximum amplitude at 1.5 Hz due to resonance [1]

